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4th Sem / Mechanical Engg.
Subject : Thermodynamics - II

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1 Crank case ventilation is the process of removing
- a) Water vapours b) Fuel vapours
 - c) Combustion gases d) None of the above
- Q.2 In a diesel engine, thermal efficiency lies in the range of
- a) 30-35% b) 15-20%
 - c) 10-15% d) 80%
- Q.3 The air fuel ratio required for idling is
- a) 10:1 b) 10:1 to 11:0
 - c) 12.5:1 to 13.5:1 d) 16.5:1 to 17.5:1
- Q.4 In plunger type fuel injection pump, plunger is operated by

- a) Belt drive
- b) Gear derive
- c) cam and tappet mechanism
- d) none of above

Q.5 An engine indicator is used to find

- a) B.P.
- b) I.P.
- c) Volume
- d) temperature

Q.6 The ram jet engine has

- a) single compressor unit
- b) double compressor unit
- c) no compressor unit
- d) either (a) and (b)

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 A gas turbine is an I.C. engine.(True/False)

Q.8 Define condenser efficiency.

Q.9 Write two applications of steam nozzles.

Q.10 Give two alternative fuels for I.C. engine.

Q.11 Expand MPFI.

Q.12 What is air-fuel ratio?

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Explain closed cycle constant pressure gas turbine.

Q.14 What is jet condenser, explain any one type of them.

Q.15 Write the working principle of impulse steam turbine.

Q.16 Explain the PV and TS diagram for the Diesel cycle.

Q.17 What is the various function of lubrication.

Q.18 What is the function of nozzle, explain multi-hole nozzle.

Q.19 What is battery ignition system? Explain

Q.20 Explain the factors which affect carburetion.

Q.21 Explain any one commercial tests of IC engine.

Q.22 What is turbo-jet engine with its principle.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 What is cooling system? Explain in details the type of cooling system with diagram and application.

Q.24 a) Write a short note on governing of steam turbine.

b) What is the working of turbocharger?

Q.25 What is carburetor? Explain in details the working of Solex carburetor.